When is Britain’s electricity cleanest, and can we trust the forecast?

90 days of grid data, 8 August to 5 November 2026 — 4,320 half-hourly readings

The short version

Electricity in Great Britain is roughly **twice as clean at lunchtime as at teatime**. Over the last 90 days the average half-hour at midday carried 86 gCO_2/kWh; the average half-hour at 7pm carried 156. Anything that can be moved — charging, heating water, a batch job, a wash — is worth moving into the middle of the day. The published forecast is good enough to plan against: within 20 grams of the real number in **84% of half-hours**, and within 50 grams in **97%**.

When the grid is greenest

Time of day	Average intensity (gCO_2/kWh)
Afternoon, 12:00–17:00	**100**
Morning, 06:00–12:00	110
Overnight, 00:00–06:00	116
Late evening, 21:00–24:00	131
Evening peak, 17:00–21:00	**151**

The cleanest single half-hour is **12:00–12:30** at 86; the dirtiest is **19:00–19:30** at 156. An 81% swing within one day, repeating almost every day.

- **Weekends are cleaner than weekdays** — 104 against 125, about 16% lower. Demand is lower, so less gas generation is called on.
- **Overnight is not the green option it is assumed to be.** It beats the evening but loses to the afternoon. Advice to “run it overnight” is out of date on the current grid.
- Across the 90 days, **36% of half-hours were rated low** carbon by the official index, 43% moderate and 21% high or very high.

How reliable the forecast is

- **Average error 12 gCO_2/kWh**, about 10% of a typical reading. Within 20 grams 84% of the time, within 50 grams 97%.
- **Slight optimism: +2.8 grams.** The forecast says the grid will be marginally cleaner than it turns out to be — small, but consistent across every month in the window.
- **Worst single half-hour: 165 grams out.** Large errors are rare and cluster around rapid changes in wind output.

In practice: trust it for *choosing a time of day*; use the settled actual for anything reported or billed.

What this does not tell you

1. **Ninety days is one season.** Late summer into early autumn. November already looks worse than September (165 against 108), and no winter is covered.
2. **It is a national average.** Scotland and the south of England differ considerably; a regional endpoint exists and is not used here.
3. **There is no weather in the model.** Wind drives most of the variation, and nothing here explains *why* a day was clean or dirty.
4. **The most recent hours are forecast only.** Six of 4,320 periods had no settled actual when loaded; they are left empty rather than estimated.

If we acted on this

Narrow and defensible: **shift discretionary electricity use out of 17:00–21:00 and into 11:00–14:00.** On these numbers that cuts the carbon intensity of whatever is moved by roughly 45%, with no reduction in how much is used.

Source: Carbon Intensity API, National Energy System Operator, licensed CC BY 4.0. Figures from the pipeline run of 5 November 2026 covering 8 August – 5 November 2026. Full method, code and data-quality results in the project repository.